Hypothesis Testing I

Unit - 4

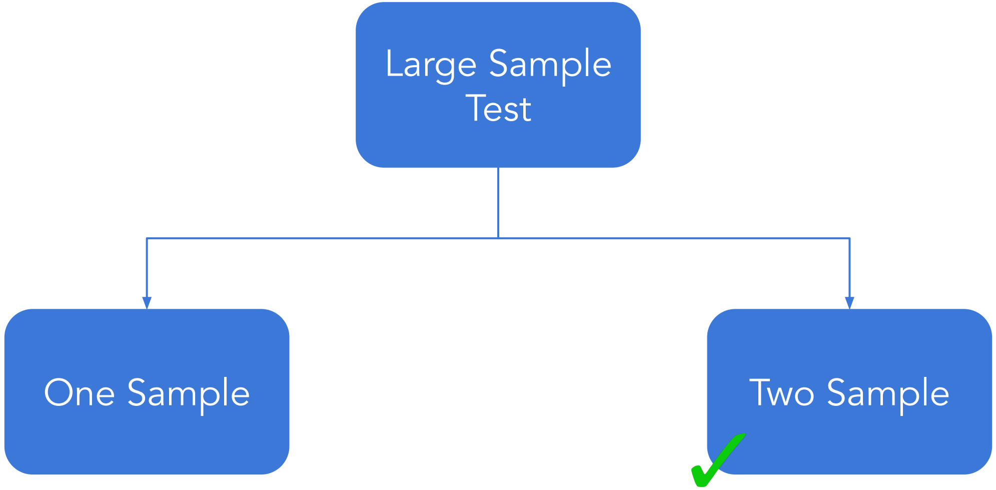
Session - 8

Unit 4: Hypothesis Testing

- Test of Mean - One sample Z Test - Test of Mean
- One sample t-Test - Test of Mean
- 2 samples - Independent and Dependent - 2 samples
- Test of proportion - One sample (Z proportion test)
- Two samples (Z proportion test)
- Case Study - Categorical vs Continuous
- More than 2 samples

Two Sample Test

Large sample tests



Large sample tests

- One sample
- Two sample

Number of Defective Items from Production Line A	Number of Defective Items from Production Line B
1	5
3	8
5	2
6	5
2	3
4	6

Can we say that Production Line B produces less defectives than Production Line A?

Two sample tests

- The two sample tests are used to compare the equality of means of two populations

- Used to compare:
  - Performance of two machineries
  - Performance of two portfolios

Two sample tests for population mean

- Let there be two different populations such that they follow normal distribution
- The two samples are independent of each other
- The samples must have equal variance (it can be tested statistically)

PLEASE!! NOTE

Test for equality of variances

- The hypothesis to test whether two populations have equal variances

H_0: The variances are equal   against H_1: The variances are not equal

- Failing to reject H_0, implies that the two populations have equal variance

- The Levene's test used

Note: The python code for levene’s test scipy.stats.levene()

Two sample test - hypothesis

- Let there be two samples of sizes n_1 and n_2 drawn from normal population $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively

- The hypothesis to test whether the population means are equal

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_1$$

- It implies

H_0: The two population means are equal (i.e $\mu_1 = \mu_2$)

against H_1: The two population means are not equal μ_0 (i.e $\mu_1 \neq \mu_2$)

Two sample test - hypothesis

- Like the one sample tests, it is possible to test for

$$H_0 : \mu_1 \leq \mu_2 \quad \text{against} \quad H_1 : \mu_1 > \mu_2$$

Or

$$H_0 : \mu_1 \geq \mu_2 \quad \text{against} \quad H_1 : \mu_1 < \mu_2$$

- Failing to reject H_0 implies that the null hypothesis is true

Two sample tests - test statistic

- The test statistic is Z given by

Sample mean

Specified mean

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- Under H_0, the test statistic follows normal distribution

Two sample tests - test statistic

- If σ_1^2 and σ_2^2 are not known then they are replaced the sample estimates s_1^2 and s_2^2 respectively

- The test statistic is Z given by

Sample mean

Specified mean

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

where $s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$

PLEASE!! NOTE

The python code to conduct a Z test for two population means is

```
statsmodels.stats.weightstats.ztest(Sample_1, Sample_2, value, alternative)
```

Two Sample Test (σ known)

Compare two population parameters such as mean

1) Difference in two population Means when σ is known

Assumptions:

1. Two samples of sizes n_1, n_2 are drawn from two populations. n_1 and n_2 are large (at least 30)
2. The samples are drawn from two populations with means μ_1 and μ_2 and corresponding standard deviations, σ_1 and σ_2 are known.

Let $\bar{x}_1$ and $\bar{x}_2$ are the estimated mean values from two samples from the two populations. The statistic $(\bar{x}_1 - \bar{x}_2)$ follows a standard normal distribution with mean $(\mu_1 - \mu_2)$ and standard deviation $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$

$$z = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Two-Sample test - Z test for Mean (σ known)

Example: Salary of Management trainees with MBA from Premier institute and other institutes are given below:

Assume that the salary of Management trainees with MBA from Premier and non-Premier institutes follow normal distribution.
Check whether the difference in monthly salary is more than 5000 for Management trainees with MBA from Premier Institutes.

Type of Management Institute	Sample Size	Mean monthly salary (INR)	Population standard deviation
Premier	120	67500	7200
Non-Premier	45	58950	4600

Two-Sample test - Z test for Mean (σ known)

Solution

We observe from the data,

$n1 = 120$ and $n2 = 45$ $\qquad \overline{X_1} = 67500$, $\overline{X_2} = 58950$

$\sigma_1 = 7200; \sigma 2 = 4600$

Step 1: State the null and alternative hypothesis

$H_o: \mu 1 - \mu 2 \leq 5000$

$H_a: \mu 1 - \mu 2 > 5000$

Step 2: Choose the level of significance, α to be 5%.

Step 3: Choose the appropriate test statistic. Since σ is known, you use the Z distribution and Z as test statistic.

Two-Sample test - Z test for Mean (σ known)

Step 4: Determine the critical value that divides the rejection and non-region regions.

The alternative hypothesis has > sign and therefore has one **tail**. The area of rejection of the Z distribution's right (upper) tail is 0.05. The Z critical value is 1.64 at 5 % level of significance.

Using Critical Value approach

Step 5: Collect the data, organize the results and compute the value of the test statistic.

$$z = \frac{((67500 - 58950) - 5000)}{\sqrt{\frac{7200^2}{120} + \frac{4600^2}{45}}} = 3.737$$

Step 6: State the statistical decision and the managerial conclusion.

- Z is in the region of rejection because $Z > Z_{CRITICAL}$ (= 1.64) where Z = 3.7374
- So, the statistical decision is to reject the null hypothesis.
- The managerial conclusion is that sufficient evidence exists to prove that difference in monthly salary is at least 5000 or more for Management trainees with MBA from Premier Institutes than others.

Two-Sample test - Z test for Mean (σ known)

Step 6 : Determine the critical value that divides the rejection and non-region regions.

The alternative hypothesis has > sign and therefore has one **tail**. The area of rejection of the Z distribution right (upper) tail is 0.05. The Z critical value is 1.64 at 5 % level of significance.

P Value approach

Step 7: Collect the data, organize the results and compute the value of the test statistic.

P-value = 9.296022e-05

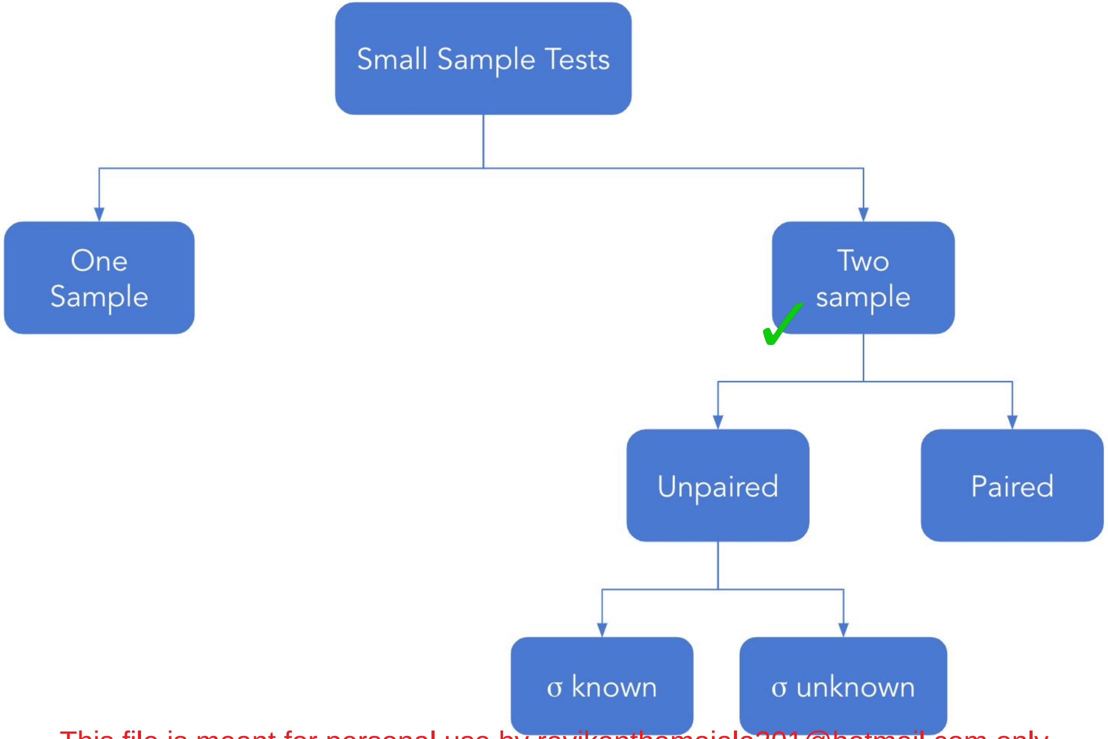
Step 8: State the statistical decision and the managerial conclusion.

- Since P-value (= 9.296022e-05) < 0.05 (our level of significance), we reject null hypothesis.
- The managerial conclusion is that sufficient evidence exists to prove that difference in monthly salary is at least 5000 or more for Management trainees with MBA from Premier Institutes than others.

Two sample tests - decision rule

<table>
<tr><th></th><th>H_1</th><th>Based on critical region</th><th>Based on p-value</th><th>Based on confidence interval</th></tr>
<tr><td>For two tailed test</td><td>$\mu_1 \neq \mu_2$</td><td>Reject H_0 if $|Z| \geq Z_{\alpha/2}$</td><td rowspan="3">Reject H_0 if p-value is less than or equal to the level of significance</td><td rowspan="3">Reject H_0 if $\mu_1 - \mu_2$ does not lie in the confidence interval</td></tr>
<tr><td>For left tailed test</td><td>$\mu_1 < \mu_2$</td><td>Reject H_0 if $Z \leq Z_\alpha$</td></tr>
<tr><td>For right tailed test</td><td>$\mu_1 > \mu_2$</td><td>Reject H_0 if $Z \geq Z_\alpha$</td></tr>
</table>

Small sample tests



t Test - 2 Sample

Two sample tests

- Like the two sample Z test, the two sample t test are used to compare the equality of means of two populations for unpaired data

- For unpaired data, test relative performance two machineries, investment portfolios

Two sample tests for unpaired data

- To test population means of two populations
- Let there be two different populations such that they follow normal distribution
- Two samples of sizes n_1 and n_2 drawn from normal population $N(\mu_1, \sigma_1^2)$ and $N(\mu_2, \sigma_2^2)$ respectively
- The two samples are independent of each other

Two sample test - hypothesis

- The hypothesis to test whether the population means are equal

$$H_0 : \mu_1 = \mu_2 \quad \text{against} \quad H_1 : \mu_1 \neq \mu_1$$

- It implies

H_0: The two population means are equal (i.e $\mu_1 = \mu_2$)

against H_1: The two population means are not equal μ_0 (i.e $\mu_1 \neq \mu_2$)

- The one sided hypothesis are

$H_0 : \mu_1 \leq \mu_2$ against $H_1 : \mu_1 > \mu_2$ Or $H_0 : \mu_1 \geq \mu_2$ against $H_1 : \mu_1 < \mu_2$

Two sample tests - test statistic

- The test statistic is Z given by

σ_1^2 and σ_2^2 are known

$$Z = \frac{(\bar{X} - \bar{Y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Two sample tests - test statistic

- The test statistic is Z given by

σ_1^2 and σ_2^2 are known

$$Z = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1}+\frac{\sigma_2^2}{n_2}}}$$

σ_1^2 and σ_2^2 are unknown

$$t = \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{s\sqrt{\frac{1}{n_1}+\frac{1}{n_2}}}$$

Where s^2 is the pooled sample variance given by $s^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}$

- If σ_1^2 and σ_2^2 are not known then they are replaced the sample estimates s_1^2 and s_2^2 respectively

Two sample tests - test statistic

- If σ_1^2 and σ_2^2 are not known then they are replaced the sample estimates s_1^2 and s_2^2 respectively
- The pooled sample variance (s^2) is used
- Under H_0, the test statistic follows t distribution with $n_1 + n_2 - 2$ df
- Failing to reject H_0 implies that the two population means are equal (i.e $\mu_1 = \mu_2$)

PLEASE!! NOTE

The python code to conduct a t test for two population means which are not paired is

```
scipy.stats.ttest_ind(Sample_1, Sample_2)
```

Two sample test for population mean (σ unknown)

Question:

A study was carried out to understand amount of hemoglobin in blood for males and females. A random sample of 160 males and 180 females have means of 13 g/dl and 15 g/dl. The two samples have standard deviation of 4.1 g/dl for male donors and 3.5 g/dl for female donor . Can it be said the population means of hemoglobin are the same for men and women? Use $\alpha = 0.01$.

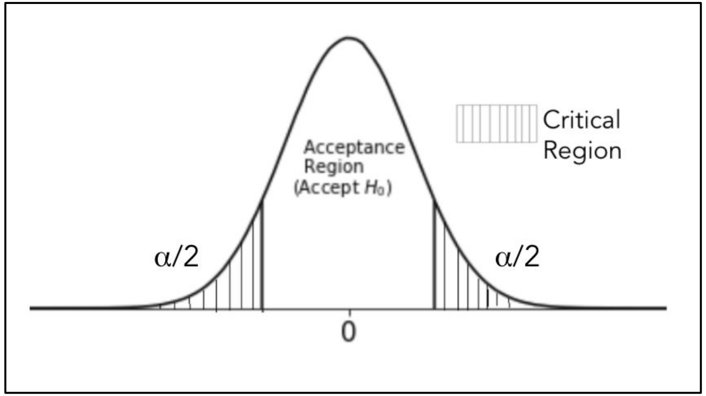
Two sample test for population mean (σ unknown)

Solution:

X: Amount of hemoglobin in blood for males
Y: Amount of hemoglobin in blood for females

Here $n_1 = 160$, $s_1 = 4.1$, $\bar{X} = 13$
$n_2 = 180$, $s_2 = 3.5$, $\bar{Y} = 15$

To test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$



Two sample test for population mean (σ unknown)

Solution:

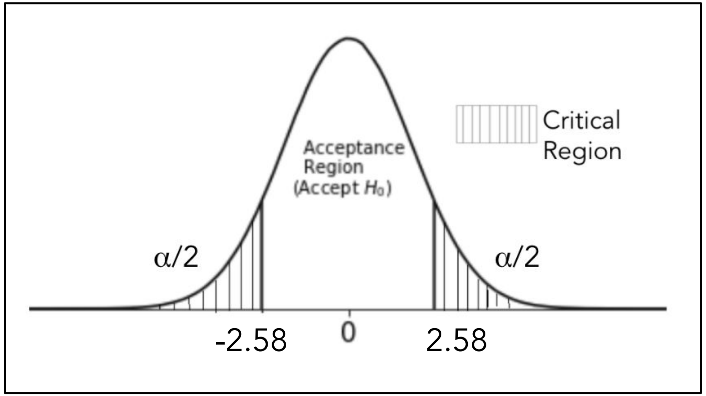
The test statistic $= \frac{(\bar{X}-\bar{Y})-(\mu_1-\mu_2)}{\sqrt{\frac{s_1^2}{n_1}+\frac{s_2^2}{n_2}}} = \frac{13-15}{\sqrt{\frac{4.1^2}{160}+\frac{3.5^2}{180}}} = -4.807$

Decision Rule: Reject $|t_{calc}| \geq t_{\alpha/2}$

Here $t_{\alpha/2} = 2.58$

Since $4.807 > 2.58$, reject H_0.

We may conclude that both males and females have different hemoglobin averages



Two sample test for population mean (σ unknown)

solution: Calculate critical t-value

i.e. If test statistic is less than -2.58 or greater than 2.58

then we reject H_0.

Two sample test for population mean (σ unknown)

solution: Calculate test statistic

As test statistic (-4.8068) < critical value (-2.58),

we reject H_0.

Two sample test for population mean (σ unknown)

solution: Calculate p-value

As p-value < 0.01, we reject H_0

Two sample test for unpaired data

Question:

An experiment was conducted to compare the pain relieving hours of two new medicines. Two groups of 14 and 15 patients were selected and were given comparable doses. Group 1 was given medicine 1 and group 2 was given medicine 2. Following data is obtained from the two samples. Test whether the two populations give the same mean hours of relief. Assume the data comes from normal distribution has equal variance. [Use $\alpha = 0.01$]

	Medicine 1	Medicine 2
Mean of hours of relief	6.4	7.3
S.D of hours of relief	1.4	1.5

Two sample test for unpaired data

Solution:

Let X: patients receiving medicine 1
Y: patients receiving medicine 2

To test $H_0: \mu_1 = \mu_2$ against $H_1: \mu_1 \neq \mu_2$

The pooled variance is

$$s^2 = \frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2} = \frac{(13)1.4^2+(14)1.5^2}{14+15-2} = 2.11$$

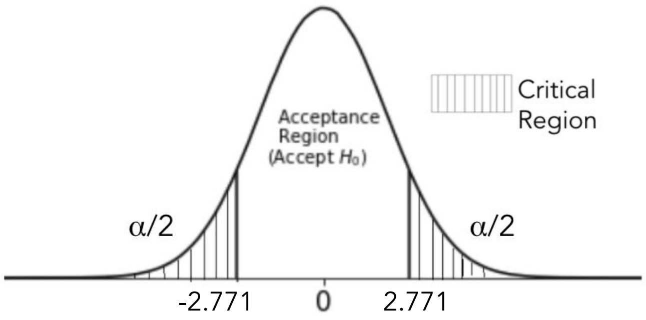
Two sample test for unpaired data

Solution:

The test statistic is $$t = \frac{(\bar{X} - \bar{Y})}{s\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} = \frac{(6.4 - 7.3)}{1.4525\sqrt{\frac{1}{14} + \frac{1}{15}}} = -5.69$$

Decision rule: Reject H_0 if $|t_{cal}| \geq t_{n1+n2-2,\, \alpha/2}$

$t_{n1+n2-2,\, \alpha/2} = t_{27,0.005} = 2.771$



Since 2.771>|-1.667|, we fail to reject H_0.

The two medicines have the same mean hours of sleep.

Two sample test for unpaired data

Python solution: Calculate critical t-value

```
# calculate the t-value for 99% of confidence level
# use 'stats.t.isf()' to find the t-value corresponding to the upper tail probability 'q'
# pass the value of 'alpha/2' to the parameter 'q', here alpha = 0.01
# pass the degrees of freedom to the parameter, 'df'
# use 'round()' to round-off the value to 2 digits
t_val = np.abs(round(stats.t.isf(q = 0.01/2, df = 27), 2))

print('Critical value for two-tailed t-test:', t_val)
```

```
Critical value for two-tailed t-test: 2.77
```

If test statistic is less than -2.77 or greater than 2.77, then we reject H_0.

Two sample test for unpaired data

Python solution: Calculate test statistic

```
# size of first sample
n_1 = 14

# mean hours for medicine 1
samp_avg_1 = 6.4

# standard deviation of hours for medicine 1
samp_std_1 = 1.4

# size of second sample
n_2 = 15

# mean hours for medicine 2
samp_avg_2 = 7.3

# standard deviation of hours for medicine 2
samp_std_2 = 1.5

# calculate pooled standard deviation
s = np.sqrt((((n_1-1)*samp_std_1**2) + ((n_2-1)*samp_std_2**2)) / (n_1 + n_2 - 2))

# calculate the test statistic
t_stat = (samp_avg_1 - samp_avg_2) / (s * np.sqrt(1/n_1 + 1/n_2))

# print the test statistic value
print('Test Statistic:', t_stat)
```

```
Test Statistic: -1.667146701465023
```

As test statistic (=-1.66) > critical value (=-2.59), we fail to reject H_0.

Two sample test for unpaired data

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'cdf()' to calculate P(t <= t_stat)
# pass the degrees of freedom to the parameter, 'df'
p_value = stats.t.cdf(t_stat, df = n_1 + n_2 - 2)

# for a two-tailed test multiply the p-value by 2
req_p = p_value*2
print('p-value:', req_p)
```

```
p-value: 0.10704510910964088
```

As p-value > 0.01, we fail to reject H_0.

Testing for Mean Difference of Two Population Independent Samples – Small Sample Case

- Let $\bar{x}_1 - \bar{x}_2$ be the population mean difference .
- In this case we use t statistic.
- Assuming equal variances for both populations
  - If population σ is known, use it
  - Otherwise we use pooled variance given by
  - $s^2 = \frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{(n_1 + n_2 - 2)}$
  - Sample SD $\quad s^2_{(\bar{x}_1 - \bar{x}_2)} = \sqrt{s^2\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$

Independent Sample t Test

An aptitude test was conducted in which groups of engineering and accounting executives participated. Is there a significant difference in the performance the groups?

Scores									
Engineers	125	115	119	85	97	107	125	125	118
Accountants	112	98	109	96	77	70	114	100	

	Formula	Engineers	Accountants
Mean		112.889	97
Variance		196.611	256.857
Observations		9	8

Independent Sample t Test

SRM INSTITUTE OF SCIENCE & TECHNOLOGY
(Deemed to be University u/s 3 of UGC Act, 1956)

Solution:

t-Test: Two-Sample Assuming Equal Variances		
	Engineers	Accountants
Mean	112.889	97
Variance	196.611	256.857
Observations	9	8
Pooled Variance	224.726	
Hypothesized Mean Difference	0	
df	15	
t Stat	2.181	
P(T<=t) one-tail	0.023	
t Critical one-tail	1.753	
P(T<=t) two-tail	0.045	
t Critical two-tail	2.131	

This is a 2 Tailed Test. Since positive $t > t_{critical}$, we reject null hypothesis. There is significant difference is performance.

- The test is carried out under the assumption that the samples are drawn from independent normal population

- If the populations from which the samples are drawn do not have equal variance then the t-test can not be use. In this case the Welch test is used

Matched / Paired Sample Hypothesis Test

- Compare a population parameter such as mean before and after intervention.
- In a paired t test, the data related to a parameter is captured twice, once before the intervention and once after the intervention. (Example: Amount of alcohol consumed by people before and after marriage break-up)

Assumptions:

The differences between the estimated parameter before and after intervention follow a normal distribution.

- Test statistic, $t = (D - \mu_D) / (S_D / \sqrt{(n)})$ follows a t distribution with n -1 degrees of freedom.
- Here D is the mean difference in the estimated parameter values before and after the treatment
- μ_D is the hypothesized mean difference and S_D is the corresponding standard deviation.

Two sample tests for paired data

- For paired data effectiveness of some training/treatment is measured

- The observations are recorded on the same individual/item twice resulting in pairs of observations

Example:

A energy drink manufacturing company wants to test if sales increase after they advertise the drink with a life sized picture of a well-known athlete.

Two sample tests for paired data

- Let $\{(X_i, Y_i), i = 1, 2, 3, \ldots, n\}$ where X and Y are paired data
- Let μ_X, μ_Y be the mean of data from X and Y respectively
- Define $d_i = y_i - x_i$
- Let $\mu_d = \mu_y - \mu_x$
- In paired t-test, we test for

$$H_0 : \mu_d = \mu_0 \quad \text{against} \quad H_1 : \mu_d \neq \mu_0$$

$$H_0 : \mu_d \leq \mu_0 \quad \text{against} \quad H_1 : \mu_d > \mu_0$$

$$H_0 : \mu_d \geq \mu_0 \quad \text{against} \quad H_1 : \mu_d < \mu_0$$

Two sample tests for paired data

- Suppose mean of d_i = $\bar{d} = \frac{\sum d_i}{n}$ and variance of d_i = $s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1}$
- The test statistic is

$$t = \frac{\bar{d} - \mu_d}{s/\sqrt{n}}$$

- Under H_0, the test statistic follows t-distribution with n-1 degrees of freedom
- Failing to reject H_0, implies that the null hypothesis is true

The python code to conduct a t test for two population means which are paired is

```
scipy.stats.ttest_rel(Sample_1, Sample_2)
```

Two sample t-test for paired data

Question:

An energy drink distributor claims that a new advertisement poster, featuring a life-size picture of a well-known athlete. For a random sample of 10 outlets, the following data was collected. Test that the null hypothesis that there the advertisement was effective in increasing the sales

Before	33	32	38	45	37	47	48	41	45
After	42	35	31	41	37	36	49	49	48

Test the hypothesis using critical region technique. [Use $\alpha = 0.05$].

Two sample t-test for paired data

Solution:

To test,

H_0: The advertisement was not effective ($\mu_d \leq 0$) against H_1: The advertisement was effective ($\mu_d > 0$)

Compute d_i

Before	33	32	38	45	37	47	48	41	45
After	42	35	31	41	37	36	49	49	48
$d_i=y_i-x_i$	9	3	-7	-4	0	-11	1	8	3

Two sample t-test for paired data

Solution:

To test,

H_0: The advertisement was not effective against H_1: The advertisement was effective

($\mu_d \le 0$) ($\mu_d > 0$)

We have mean $\bar{d} = \frac{\sum d_i}{n} = 0.22$ and its variance $s^2 = \frac{\sum (d_i - \bar{d})^2}{n-1} = 43.6944$

The test statistic is

$$t = \frac{d - \mu_d}{s/\sqrt{n}} = \frac{0.22 - 0}{6.6101/\sqrt{9}} = \quad 0.1008$$

Two sample t-test for paired data

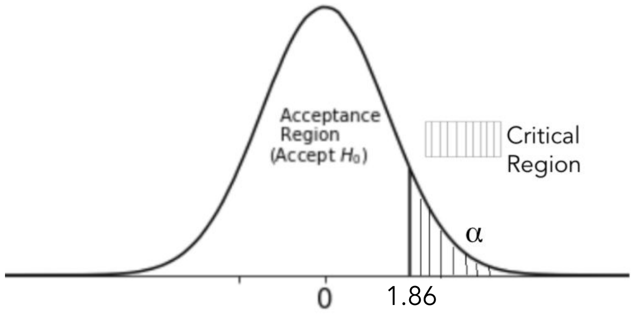
Solution:

Decision Rule: Reject H_0, if $t_{calc} \geq t_{df,\alpha}$

Here $t_{df,\alpha} = t_{n-1,\alpha} = t_{8,0.05} = 1.86$

Since 1.86 > 0.0998, fail to reject H_0

Thus, there is no effect of the advertisement.



Two sample t-test for paired data

Python solution: Calculate critical t-value

```
# calculate the t-value for 95% of confidence level
# use 'stats.t.isf()' to find the t-value corresponding to the upper tail probability 'q'
# pass the value of 'alpha' to the parameter 'q', here alpha = 0.05
# pass the degrees of freedom to the parameter, 'df'
# use 'round()' to round-off the value to 2 digits
t_val = round(stats.t.isf(q = 0.05, df = 8), 2)

print('Critical Value for one-tailed t-test:', t_val)
```

```
Critical Value for one-tailed t-test: 1.86
```

If test statistic is greater than 1.86, then we reject H_0.

Two sample t-test for paired data

Python solution: Calculate test statistic and p-value

```
# use 'ttest_rel()' to calculate the t-statistic and corresponding p-value for paired samples
# pass the after and before sales to the function
t_stat, p_val = stats.ttest_rel(sales_after, sales_before)

# print the t-test statistic and corresponding p-value
print("Test Statistic:", t_stat)
print("p-value:", p_val)
```

The test statistic (=0.1008) < critical value (=1.86), also the p-value > 0.05, thus we fail to reject H_0.

Example of Paired Sample t Test

A researcher believes that Cholesterol level of participants of a meditation program will be lesser after completing the program than before attending the program. Check the claim at 5% level of significance.

Sample size = 50, D = 14 S_D = 8.5 μ_D = 10 mg/dL

Solution:

Step 1: State the null and alternative hypothesis

Ho: $\mu_D \leq 10$
Ha: $\mu_D > 10$

Step 2: Choose the level of significance, α to be 5% and sample size is 50

Step 3: Choose the appropriate test statistic. Since σ is unknown, you use the t distribution and t as test statistic.

Paired Sample t Test

Using P Value approach

Step 4: Determine the critical value that divides the rejection and non-rejection regions.

n=50, D = 14 S_D = 8.5 μ_D = 10 mg/dL

$$t = \frac{(D - \mu_D)}{\left({s_D}/{\sqrt{n}}\right)} = \frac{(14 - 10)}{{8.5}/{\sqrt{50}}} = 3.328$$

Critical value for t when α = 0.05 and degrees of freedom = n − 1 = 49 is 1.677

t = 3.328 with p value is .0008

Paired Two-Sample t test

Step 5: State the statistical decision and the managerial conclusion.

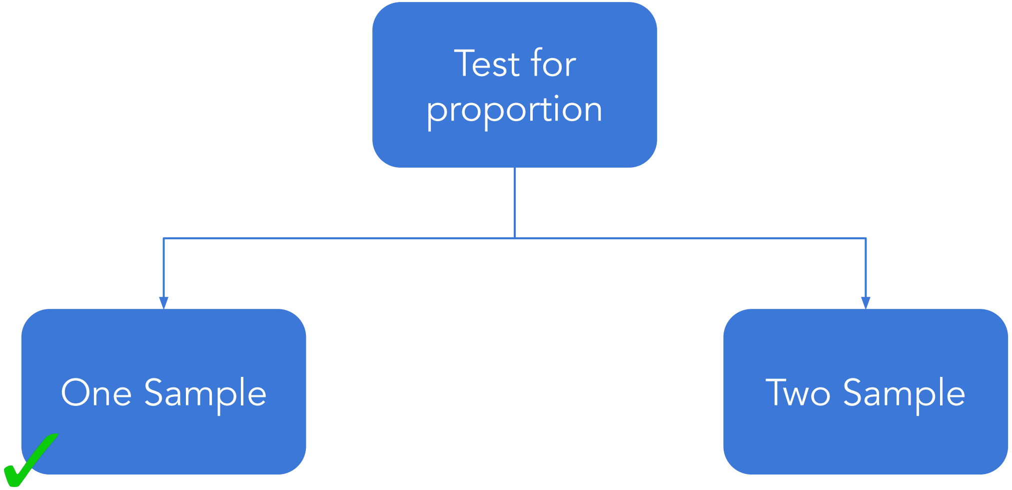
- t is in the region of rejection because p value (= 0.0008) < 0.05
- So, the statistical decision is to reject the null hypothesis.
- The managerial decision is that sufficient evidence exists to prove that cholesterol level of participants of meditation program reduces at least by 10 mg/dL after attending the meditation program at 5% level of significance.

Test for Population Proportion

Test for proportion

- For qualitative data the proportion of a desired characteristic is obtained
- Test for proportion:
  - One sample: Testing population proportion (P) is equal to a specified value (P_0)
  - Two sample: Testing equality of Two population proportions ($P_1 = P_2$)
- Similar to the tests of population mean

Test for proportion



One sample test - hypothesis

- The hypothesis to test the population proportion is equal to a specified value

$$H_0 : P = P_0 \quad \text{against} \quad H_1 : P \neq P_0$$

- It implies

H_0: The population proportion is equal to P_0

against H_1: The population proportion is not equal to P_0

- Failing to reject H_0 implies that the population proportion is equal to P_0

Test for proportions

- The test statistic is given by

Sample proportion

Specified proportion

$$Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}}$$

Sample size

- Under H_0, the test statistic follows standard normal distribution

One sample test for proportion - decision rule

	H_1	Based on critical region	Based on p-value	Based on confidence interval
For two tailed test	$P \neq P_0$	Reject H_0 if $\|Z\| \geq Z_{\alpha/2}$	Reject H_0 if p-value is less than or equal to the level of significance	Reject H_0 if P_0 does not lie in the confidence interval
For left tailed test	$P < P_0$	Reject H_0 if $Z \leq -Z_\alpha$		
For right tailed test	$P > P_0$	Reject H_0 if $Z \geq Z_\alpha$		

One sample test for proportion

Question:

From a sample 361 business owners had gone into bankruptcy due to recession. On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business. Test the null hypothesis that at most 25% of all businesses had not consulted before opening the business.

Test the claim using p-value technique. [Use $\alpha = 0.05$].

One sample test for proportion

Solution:

From a sample 361 business owners had gone into bankruptcy due to recession.
i.e. $n = 361$

On taking a survey, it was found that 105 of them had not consulted any professional for managing their finance before opening the business.

Let X: business which did not consult before
$x = 105$

The sample proportion $(p) = x/n = 105/361 = 0.2909$

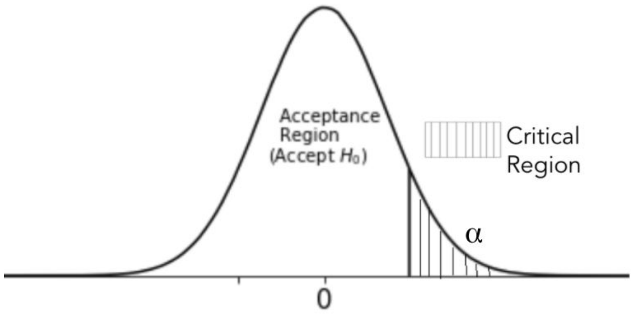
One sample test for proportion

Solution:

To test: The null hypothesis that at most 25% of all businesses had not consulted before opening the business

Here $P_0 = 0.25$

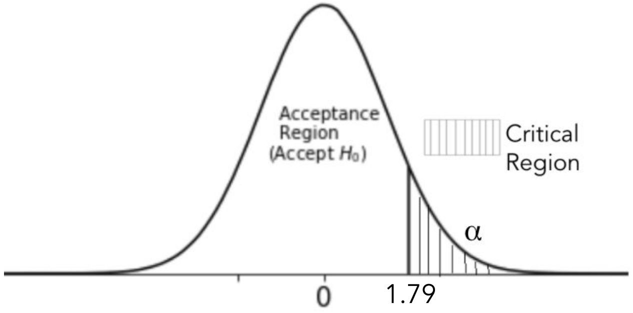
To test, H_0: $P \leq 0.25$ against H_1: $P > 0.25$



One sample test for proportion

Solution:

The test statistic $Z = \frac{p - P_0}{\sqrt{\frac{P_0(1-P_0)}{n}}} = \frac{0.2909 - 0.25}{\sqrt{\frac{0.25(0.75)}{316}}} = 1.79$



The p-value = P(Z > 1.79) = 0.0367

As p-value < 0.05, reject H_0.

We may conclude that at least 25% of all businesses had not consulted before starting the business.

One sample test for proportion

Python solution: Calculate test statistic

```
# sample size
n = 361

# number of business owners that did not consult before
x = 105

# sample proportion
p_samp = x / n

# hypothesized proportion
hypo_p = 0.25

# calculate test statistic value for 1 sample proportion test
z_prop = (p_samp - hypo_p) / np.sqrt((hypo_p * (1 - hypo_p)) / n)

print('Test statistic:', z_prop)
```

```
Test statistic: 1.7928245201151534
```

One sample test for proportion

Python solution: Calculate p-value

```
# calculate the corresponding p-value for the test statistic
# use 'sf()' to calculate P(Z > z_prop)
p_value = stats.norm.sf(z_prop)

print('p-value:', p_value)
```

```
p-value: 0.03650049373124949
```

As the p-value < 0.05, we reject H_0.

Test for proportion

When the outcome is discrete or dichotomous (success or failure), the test objective is :

Compare the proportion of successes (P-hat or P) in a single population to a known proportion (P_0)

This test will assess whether or not a sample from a population represents the true proportion from the entire population.

The proportion P-hat is calculated using the ratio of no. of successes to the sample size (n)

P-hat = x/n

Standard Deviation and Z score for Test of proportion

- Standard deviation (σ) of the sampling distribution

$$se = sqrt[P_0 * (1 - P_0) / n]$$

where P_0 is the hypothesized value of population proportion in the null hypothesis, and n is the sample size.

- Test statistic. The test statistic is a z-score (z) defined by the following equation.

$$z = (p - P_0) / se$$

where P_0 is the hypothesized value of population proportion in the null hypothesis, P is the sample proportion (sometimes referred to as P-hat), and σ is the standard deviation of the sampling distribution.

Test of proportion with One Sample - Critical Value Approach

1. State the null hypothesis H_o and the alternative hypothesis H_A.
2. Calculate the test statistic:

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where p_0 is the null hypothesized proportion i.e., when H_o: $P=P_0$

P-hat is the sample proportion

3. Determine the critical region.
4. Make a decision. Determine if the test statistic falls in the critical region. If it does, reject the null hypothesis. If it does not, do not reject the null hypothesis.

Test of proportion with One Sample - p-Value Approach

1. State the null hypothesis H_o and the alternative hypothesis H_A.
2. Calculate the test statistic:

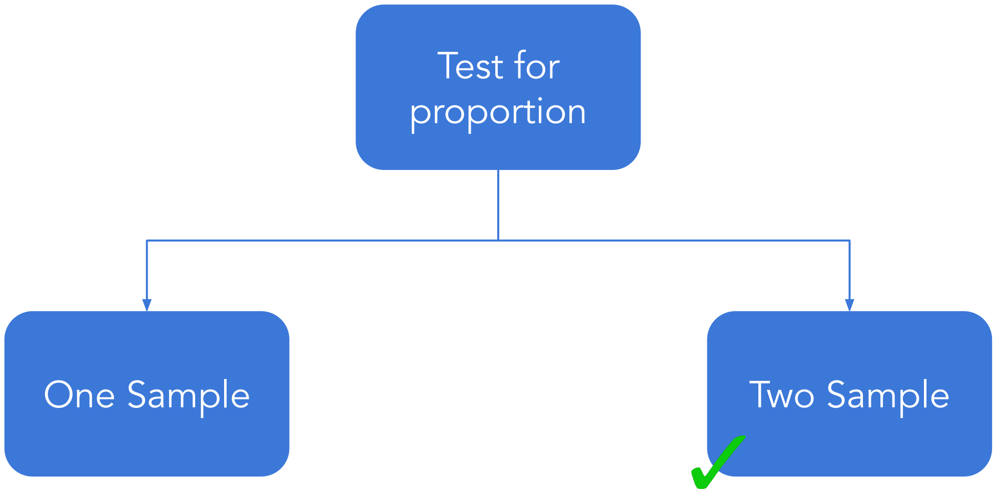
$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

where p0 is the null hypothesized proportion i.e., when $H_o: P=P_0$

P-hat is the sample proportion

3. Calculate the p-value.
4. Make a decision. Check whether to reject the null hypothesis by comparing p-value to α. If the p-value < α, then reject Ho; otherwise do not reject Ho.

Test for proportion



Test of proportion with Two Independent Samples

Objective:

Compare proportions of successes between the two groups. The relevant sample data are the sample sizes in each comparison group (n_1 and n_2) and the sample proportions (P_1-hat and P_2-hat) which are computed by taking the ratios of the numbers of successes (x_1 and x_2) to the sample sizes in each group, i.e.,

$$\hat{p}_2 = \frac{x_2}{n_2} \quad \text{and} \quad \hat{p}_1 = \frac{x_1}{n_1}$$

Test Statistic : $$z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$ where $$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}$$

Two sample tests for population proportion

- Let there be two samples sizes n_1 and n_2 from different populations of such that x_1 and x_2 are the number of specific items in each of them respectively
- Suppose these samples have proportions of specific items p_1 and p_2 respectively
- To test the equality of population proportion from which these samples are chosen

Two sample test - hypothesis

- The hypothesis to test the population proportion

$$H_0 : P_1 = P_2 \quad \text{against} \quad H_1 : P_1 \neq P_2$$

- It implies

H_0: The two population proportions are equal ($P_1 = P_2$)

against H_1: The two population proportions are not equal ($P_1 \neq P_2$)

- Failing to reject H_0 implies that the two population proportions are equal

Test for proportions

- The test statistic is given by

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1}+\frac{1}{n_2})}}$$

where $\bar{P}$ is the proportion of pooled sample such that

$$\bar{P} = \frac{x_1+x_2}{n_1+n_2} = \frac{n_1p_1+n_2p_2}{n_1+n_2}$$

- Under H_0, the test statistic follows standard normal distribution

PLEASE!! NOTE

The python code to conduct a Z test for two population proportions is

statsmodels.api.stats.proportions_ztest(Sample_1, Sample_2)

Two sample test for proportion - decision rule

<table>
<tr><th></th><th>H_1</th><th>Based on critical region</th><th>Based on p-value</th><th>Based on confidence interval</th></tr>
<tr><td>For two tailed test</td><td>$P_1 \neq P_2$</td><td>Reject H_0 if $|Z| \geq Z_{\alpha/2}$</td><td rowspan="3">Reject H_0 if p-value is less than or equal to the level of significance</td><td rowspan="3">Reject H_0 if $P_1 - P_2$ does not lie in the confidence interval</td></tr>
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</table>

Two sample test for proportion

Question:

Steve owns a kiosk where sells two magazines - A and B in a month. He buys 100 copies of magazine A out of which 78 were sold and 70 copies of magazine B out of which 65 were sold. Is there enough evidence to say that magazine is B is more popular?

Test the claim using p-value technique. [Use $\alpha = 0.05$].

Two sample test for proportion

Solution:

Steve owns a kiosk where sells two magazines - A and B in a month.

Let X: the number of magazines sold

Out of 100 copies of magazine A 78 are sold
Here, $x_1 = 78$ and $n_1 = 100$
Let p_1 be the proportion of sell of magazine A
$p_1 = x_1/n_1 = 78/100 = 0.78$

Out of 70 copies of magazine B 65 are sold
Here, $x_2 = 65$ and $n_1 = 70$
Let p_2 be the proportion of sell of magazine B
$p_2 = x_2/n_2 = 65/70 = 0.928$

Two sample test for proportion

Solution:

To test, whether magazine B is more popular, i.e

$H_0: P_1 \geq P_2$ against $H_1: P_1 < P_2$

Where P_1: denotes population proportion of magazine A sold

P_2: denotes population proportion of magazine B sold

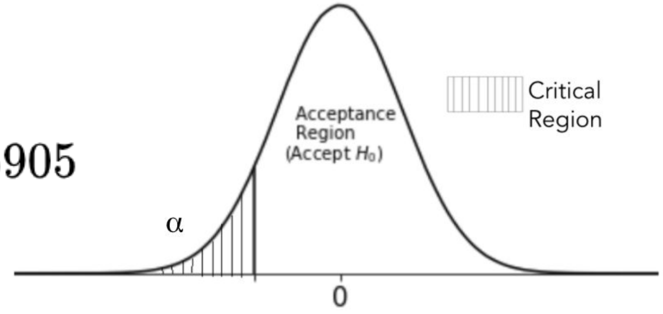
Two sample test for proportion

Solution:

The pooled proportion is $\bar{P} = \frac{x_1+x_2}{n_1+n_2} = \frac{78+65}{100+70} = 0.84$

The test statistic is

$$Z = \frac{p_1 - p_2}{\sqrt{\bar{P}(1-\bar{P})(\frac{1}{n_1}+\frac{1}{n_2})}} = \frac{0.78-0.928}{\sqrt{0.84(1-0.84)(\frac{1}{100}+\frac{1}{70})}} = -2.5905$$



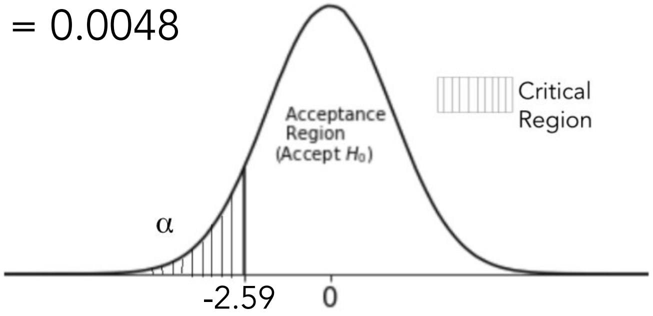
Two sample test for proportion

Solution:

The test statistic Z = -2.5905

The p-value = P(Z < Z_{calc}, under H_0) = P(Z < -2.5905, μ = 13) = 0.0048

Since p-value < 0.05, we reject H_0.



Thus there is enough evidence to conclude that magazine is B is more popular.

Two sample test for proportion

Python solution: Calculate test statistic and p-value

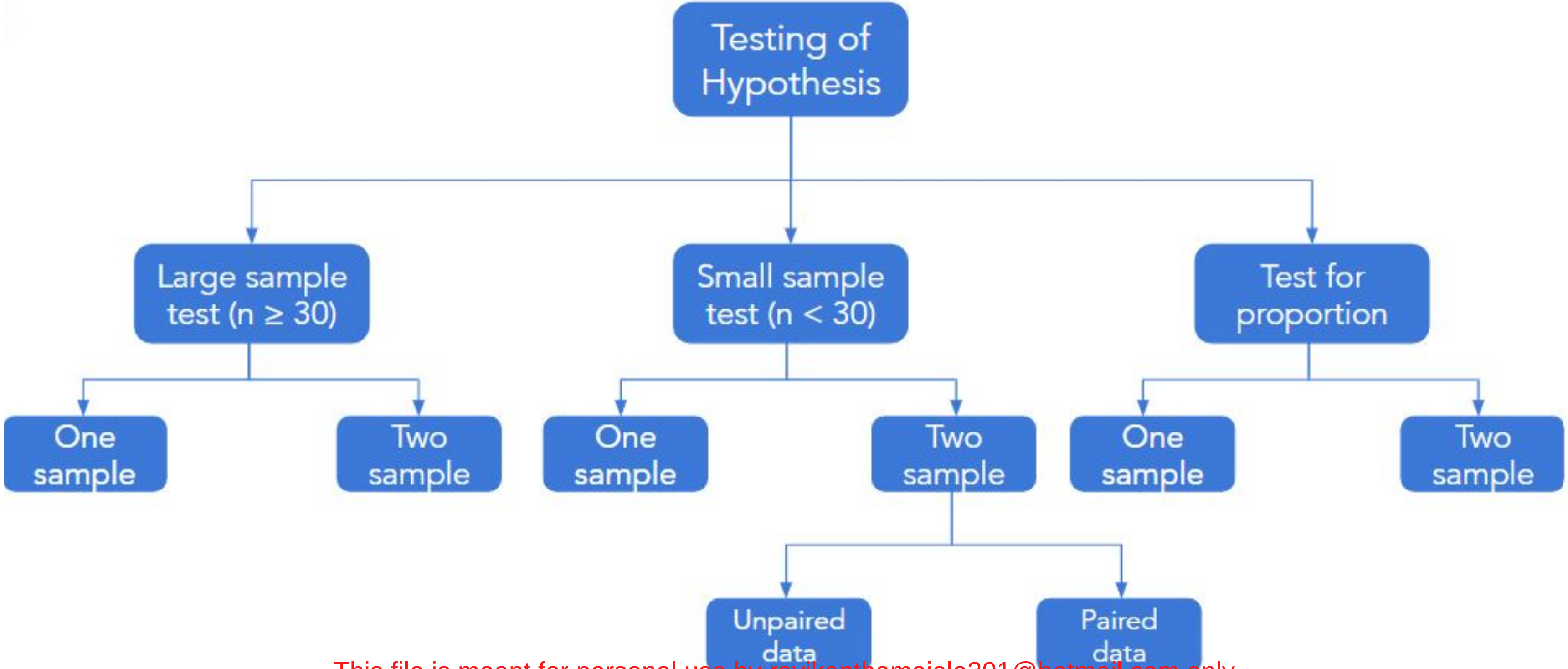
```
# calculate test statistic value for two sample proportion test
# pass the copies sold for both the magazines to the parameter, 'count'
# pass the size of both the samples to the parameter, 'nobs'
# pass the one-tailed condition to the parameter, 'alternative'
z_prop, p_val = sm.stats.proportions_ztest(count = np.array([78, 65]),
                                           nobs = np.array([100, 70]),
                                           alternative = 'smaller')

# print the value of test statistic and the corresponding p-value
print('Test statistic:', z_prop)
print('p-value:', p_val)
```

```
Test statistic: -2.60830803458311
p-value: 0.004549551600547303
```

As the p-value < 0.05, we reject H_0.

Summary



Testing of Hypothesis
Large sample test (n ≥ 30)
Small sample test (n < 30)
Test for proportion
One sample
Two sample
One sample
Two sample
One sample
Two sample
Unpaired data
Paired data

Parametric & Non parametric tests

- The tests considered so far have two features:
    - The probability distribution of the samples was assumed to be known
    - The hypothesis test was about the parameter of the probability distribution
- These tests are known as the parametric tests
- The times when these assumptions are not satisfied, use the non-parametric tests

Thank You